



Signals and systems

Course project report

(BIO 3609)

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Introduction:

This project aims to develop an Artificial Intelligence (AI) model for automated sleep stage classification using physiological sleep signals and deep learning techniques. The training pipeline is based on PSG-derived recordings, while the proposed approach is motivated by the potential of Ballistocardiography (BCG) as a future contact-free alternative for sleep monitoring.

By leveraging hybrid deep learning architectures such as CNNs and Transformers, the project focuses on improving automatic sleep stage classification performance using single-channel physiological data.

Problem Statement:

Manual sleep stage scoring is time-consuming and requires expert knowledge. Additionally, full PSG setups are expensive and intrusive.

The goal of this project is to:

- Develop an automated sleep stage classification model.
- Use a single-channel signal instead of multi-modal PSG.
- Maintain acceptable accuracy while reducing system complexity.

Single-Channel Sleep Classification. Is it possible?

Feasibility: Yes, sleep stages can be effectively classified using a single-channel signal (like ECG) by analyzing variations in the autonomic nervous system.

3. Methodology:

3.1 Data Source

The model is trained using physiological data from the Multi-Ethnic Study of Atherosclerosis (MESA) dataset. Specifically, single-channel ECG (EKG) signals were extracted from Polysomnography (PSG) recordings stored in .edf format, while corresponding sleep stage labels (Wake, N1, N2, N3, REM) were obtained from XML annotation files.

A subset of 10 patients was used in this study, resulting in a total of 12,758 segmented 30-second epochs for training and evaluation.

3.2 Preprocessing

The raw signals undergo several preprocessing steps:

- Filtering to remove noise and artifacts
- Normalization to standardize signal amplitude
- Downsampling to reduce data size
- Reshaping into fixed-length segments

3.3 Window Segmentation

The continuous signal is divided into fixed 30-second windows (epochs), which is the standard approach in sleep staging.

Each window:

- Represents a single training sample
- Is assigned one sleep stage label (Wake, N1, N2, N3, REM)

This segmentation helps:

- Standardize input size
- Capture temporal patterns
- Simplify model training

3.4 Data Preparation

- Data is split into training and testing sets
- Oversampling is applied to the N1 class (only in training) to handle class imbalance
- Data augmentation is performed by adding noise to training samples

3.5 Model Architecture

The model uses a hybrid design that combines a CNN with a Transformer encoder.

The CNN extracts meaningful patterns from the signal, while the Transformer captures temporal relationships across the sequence.

This combination allows the model to effectively learn both local features and long-term dependencies in the data.

3.6 Training Strategy

- Loss Function: Focal Loss (to handle class imbalance)
- Training performed on processed and augmented data
- Evaluation done on the original (unmodified) test set

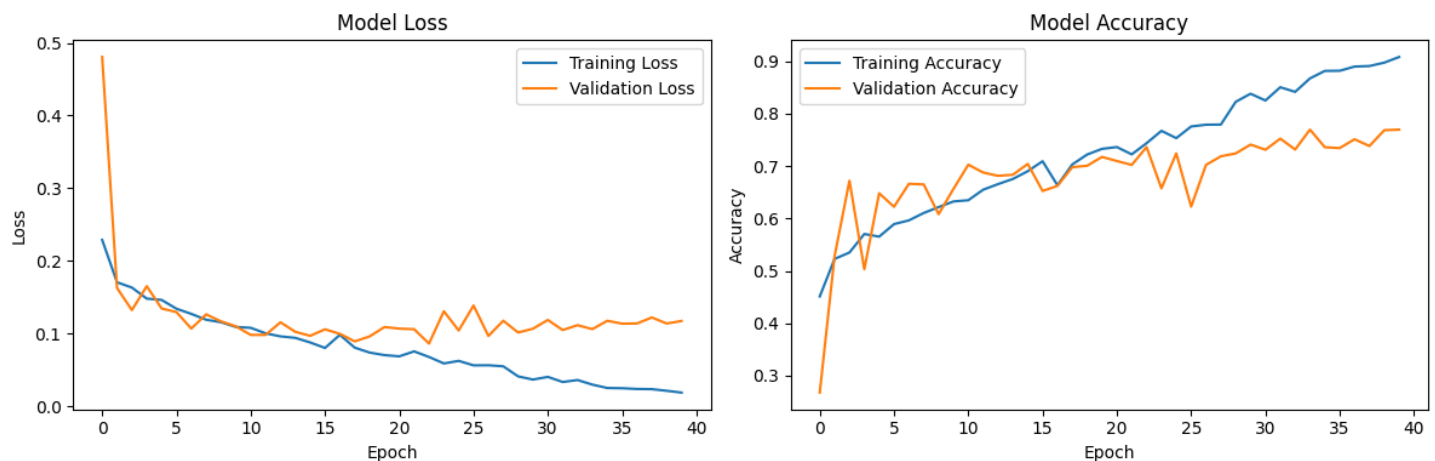
4. Results

- The proposed CNN + Transformer model achieved promising performance with Accuracy: 76.96% and Loss: 0.1060 for single-channel BCG-based sleep stage classification

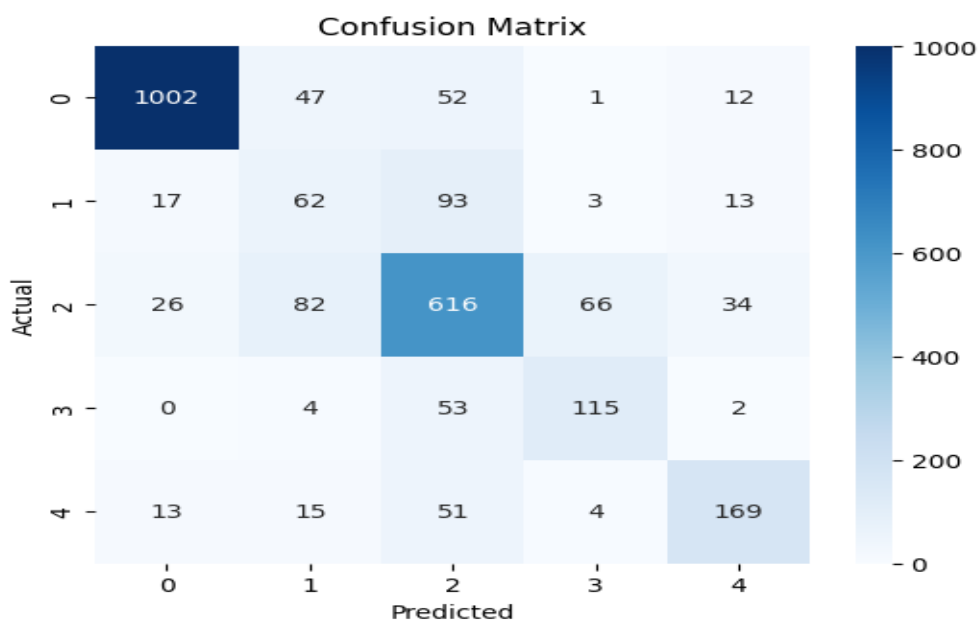
5. Evaluation

Performance analysis was conducted using:

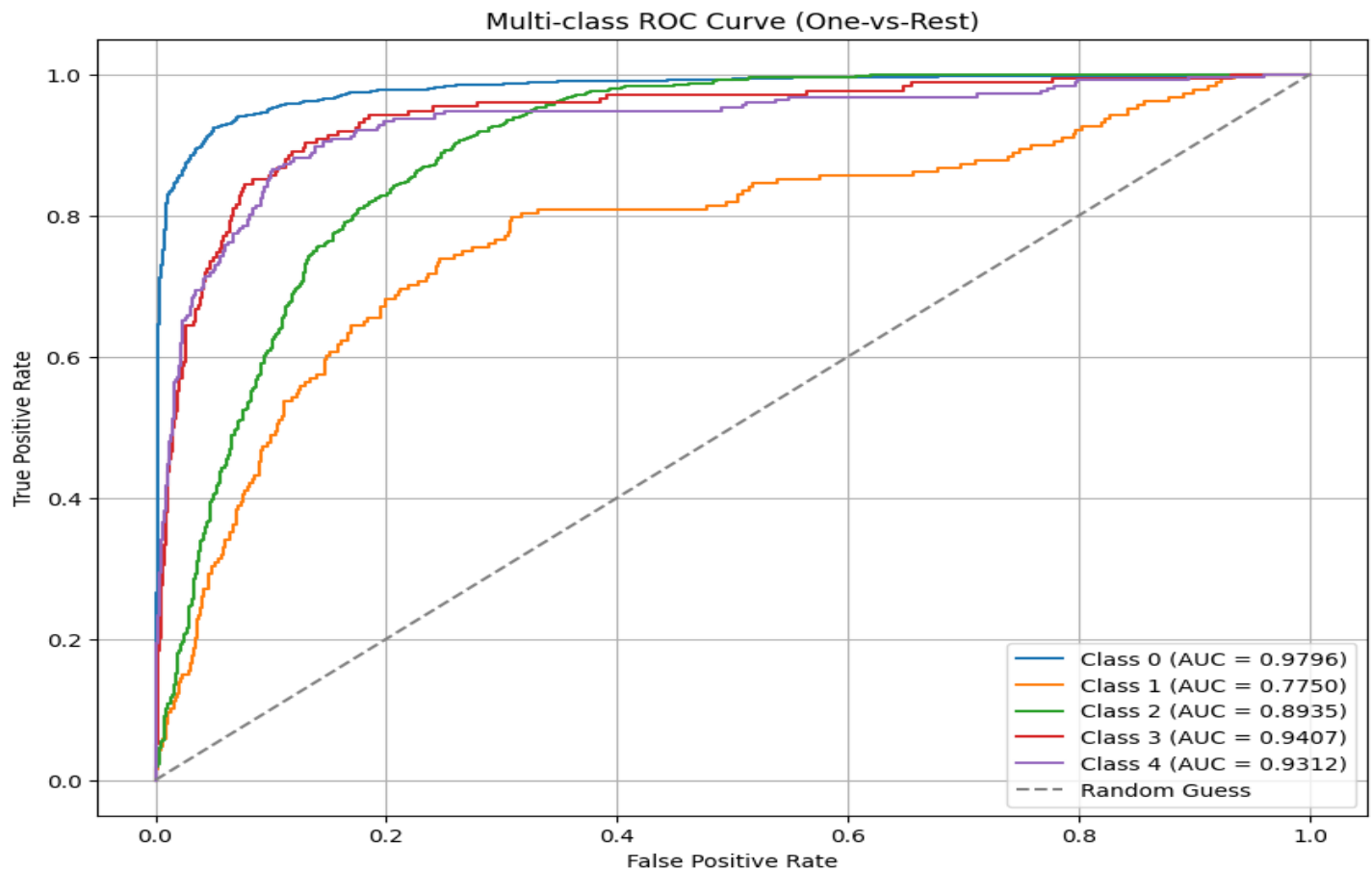
1. Accuracy and loss metrics



2. Confusion matrix visualization



3. Multi class ROC curve



5. Comparison with Related Work

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Several studies have explored automatic sleep stage classification using PSG-derived signals and deep learning models.

One study reported an overall accuracy of **60%**, highlighting the challenges of single-channel sleep classification.

Another study achieved an overall accuracy of **75.4%**, demonstrating improved performance using more advanced modeling techniques.

In addition, a deep convolutional recurrent model based on single-lead ECG signals reported an overall accuracy of approximately **74%** for five-class sleep stage classification .

In comparison, the proposed model achieved an overall accuracy of **76.96%**, slightly outperforming these approaches while maintaining a single-channel setup. These results indicate that the proposed method is competitive with existing work despite the inherent limitations of single-channel input compared to full multi-modal PSG systems.

8. Conclusion

This project presents a deep learning-based approach for sleep stage classification using single-channel physiological signals.

The model achieved competitive performance while reducing system complexity, demonstrating the feasibility of simplified sleep monitoring systems.

The work also highlights the potential of extending this approach to contact-free BCG-based systems in future research.

9. References

1-MESA-EDF Dataset

2-Relevant research papers on sleep staging and deep learning :

1. Urtnasan, E., Park, J. U., Joo, E. Y., & Lee, K. J. (2022). Deep convolutional recurrent model for automatic scoring sleep stages based on single-lead ECG signal. *Diagnostics*, 12(5), 1235. <https://doi.org/10.3390/diagnostics12051235>
2. Tsinalis, O., Matthews, P. M., Guo, Y., & Zafeiriou, S. (2016). Automatic sleep stage scoring with single-channel EEG using convolutional neural networks. arXiv preprint arXiv:1610.01683. <https://arxiv.org/abs/1610.01683>
3. Satapathy, S. K., Shah, K., Shah, S., Shah, B., & Panchal, A. (2023). A deep neural model CNN-LSTM network for automated sleep staging based on a single-channel EEG signal. In *Soft Computing for Problem Solving* (pp. 535–546). Springer, Singapore. https://doi.org/10.1007/978-981-19-6525-8_45